ORIGINAL ARTICLE

EEG-Based Analysis of Alpha Brain Activity During Eyes-Open and Eyes Closed Condition

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Abstract

Electroencephalography (EEG) is a technique used to record the electrical activity of the brain. This project investigates the differences in alpha-band power between eyes-open and eyes-closed resting conditions using an eight-channel EEG dataset.

The analysis was performed using Python. EEG signals were collected from eight channels (EXG 1–EXG 8) at a sampling frequency of 250 Hz. The signals were pre-processed using a Butterworth bandpass filter with a frequency range of 0.5–40 Hz and a 50 Hz notch filter.

Power spectral density was estimated using Welch's method. Alpha-band power was calculated within the 8–12 Hz frequency range using non-overlapping five-second windows.
The results showed that mean alpha power was higher during the eyes-closed condition than during the eyes-open condition across all eight EEG channels. The highest mean alpha power during eyes closed was observed in EXG 1, with a value of 88.276 μV^2, while the lowest was observed in EXG 6, with a value of 34.235 μV^2.

These findings demonstrate a consistent difference in alpha-band power between the two recording conditions. The observed pattern is consistent with the commonly reported increase in alpha activity during eyes-closed resting conditions. However, the analysis is based on one recording per condition, and the findings should therefore be interpreted as a descriptive comparison rather than evidence of population-level statistical significance.

KEYWORDS
Electroencephalography (EEG), Alpha Power, Eyes-Open and Eyes-Closed Conditions, Power Spectral Density, Welch's Method, EEG Signal Processing

1| INTRODUCTION

The human brain continuously produces electrical activity through the communication of neurons. This activity can be recorded using a technique known as electroencephalography (EEG). EEG measures electrical potential differences at the scalp that arise primarily from the coordinated activity of populations of neurons, providing a non-invasive method for studying brain activity (Cohen, 2017). The development of human electroencephalography can be traced to the pioneering work of Hans Berger, whose early investigations established the recording of electrical activity from the human scalp (Berger, 1929).

EEG is a non-invasive method that measures electrical signals generated by the brain through electrodes placed on the scalp. It is widely used in neuroscience and brain-computer interface (BCI) research to study brain activity under different conditions. The electrical potentials measured at the scalp are influenced by the activity of neural populations and the conduction of electrical fields through brain and surrounding tissues (Nunez & Srinivasan, 2006).

EEG signals contain different frequency components, commonly classified into frequency bands such as delta, theta, alpha, beta, and gamma. Each frequency band represents a different range of oscillatory activity. Neural oscillations are an important feature of brain activity, and their rhythmic patterns are associated with the coordination of activity across neural networks (Buzsáki, 2006).

Alpha waves generally occur within the frequency range of 8–12 Hz. Alpha activity is commonly observed during relaxed wakefulness and is often more prominent when a person closes their eyes compared with when their eyes are open (Barry et al., 2007; Klimesch, 1999). Early investigations of human electroencephalography established the presence of rhythmic electrical activity over the occipital regions of the scalp, contributing to the scientific understanding of alpha rhythms (Adrian & Matthews, 1934).

Alpha oscillations have been associated with inhibitory processes and the timing of neural activity, suggesting that alpha-band activity plays an important role in the regulation of information processing in the brain (Klimesch et al., 2007).

In this project, EEG recordings from eyes-open and eyes-closed resting conditions are analysed to investigate differences in alpha-band power across eight EEG channels. Python is used for signal preprocessing, frequency-domain analysis, alpha-power calculation, and visualisation.

The project aims to understand how alpha activity changes between the two resting conditions and to demonstrate the use of digital signal-processing techniques in EEG research.

Research Objectives

The main objectives of this project are:

1.To understand the fundamentals of Electroencephalography (EEG) and its applications in brain-computer interface research.

2.To analyse EEG signals recorded under eyes-open and eyes-closed resting conditions.

3.To preprocess EEG signals using digital filtering techniques.

4.To calculate the Power Spectral Density (PSD) using Welch's method.

5.To extract alpha-band power within the frequency range of 8–12 Hz.

6.To compare alpha power across eight EEG channels under both recording conditions.

7.To visualise and interpret the differences in alpha activity between eyes-open and eyes-closed conditions.

8.To understand the application of Python in EEG signal processing and analysis.

2 | METHOD

2.1 EEG Dataset

The EEG dataset used in this project was obtained from the Unicorn BCI Core-8 EEG Datasets repository on GitHub.

The dataset contains EEG recordings collected under different experimental conditions. For this project, two resting-state conditions were selected:

Eyes-Open Condition: EEG signals recorded while the subject's eyes were open.

Eyes-Closed Condition: EEG signals recorded while the subject's eyes were closed.

2.1.1 | EEG Recording Specifications

PARAMETER	DESCRIPTION
EEG Device	Unicorn BCI Core-8
Number of EEG Channels	8
Sampling Frequency	250 Hz
Eyes-Open Recording	1 Recording
Eyes-Closed Recording	1 Recording
Total EEG Channels Analysed	8
Alpha Frequency Range	8-12 Hz

Table 1.EEG recording specifications.

2.1.2 EEG Channels

The EEG dataset contains eight channels corresponding to different electrode locations on the scalp. These channels were used to analyse brain activity during both eyes-open and eyes-closed conditions.

The eight EEG channels used in this analysis are:

The electrode locations cover frontal, central, parietal, and parieto-occipital regions of the scalp. All eight channels were included in the analysis to compare alpha-band power between eyes-open and eyes-closed conditions.

EEG CHANNEL	ELECTRODE LOCATION
EXG 1	Fz
EXG 2	C3
EXG 3	Cz
EXG 4	C4
EXG 5	Pz
EXG 6	PO7
EXG 7	POz
EXG 8	PO8

Table 2.EEG Channels and Electrode Locations

The electrode locations cover frontal, central, parietal, and parieto-occipital regions of the scalp, following standard EEG electrode-positioning conventions (Jasper, 1958). All eight channels were included in the analysis to compare alpha-band power between eyes-open and eyes-closed conditions.

2.1.3 EEG Data Format

The EEG recordings were provided in CSV (Comma-Separated Values) format. The files contain eight EEG signal columns along with additional recording information.

The EEG data were loaded into Python using the Pandas library for further preprocessing and analysis.

2.2 Data Loading and Quality Assessment

2.2.1 Data Loading

The EEG recordings were loaded from CSV files using the Pandas library in Python. The eyes-open and eyes-closed recordings were loaded separately.

The eight EEG channels (EXG 1 to EXG 8) were selected for further analysis.

2.2.2 Data Quality Assessment

The dataset was examined to check for missing values, infinite values, and recording-quality indicators. The VALID and SATURATION columns were also inspected to assess the quality of the recorded signals.

2.3 EEG Signal Preprocessing

The EEG signals were filtered to reduce unwanted frequency components and noise. EEG preprocessing is important for improving signal quality and reducing the influence of unwanted artifacts before further analysis (Urigüen & Garcia-Zapirain, 2015).

The following filtering techniques were used:

Butterworth Bandpass Filter: A fourth-order Butterworth filter with a frequency range of 0.5–40 Hz was applied.

Notch Filter: A 50 Hz notch filter was additionally applied as part of the preprocessing pipeline. Because the preceding bandpass filter was limited to 0.5–40 Hz, the 50 Hz component was already strongly attenuated by the bandpass stage.

Zero-phase filtering was performed using forward-backward filtering functions.

2.4 Power Spectral Density Analysis

Power Spectral Density (PSD) was calculated using Welch's method to examine the distribution of signal power across different frequencies (Welch, 1967). EEG analysis requires careful consideration of signal characteristics, recording conditions, and potential sources of interference to ensure meaningful interpretation of recorded brain activity (Niedermeyer & Lopes da Silva, 2005). The following parameters describe the Welch spectral estimation procedure. The separate alpha-power summary was calculated using non-overlapping five-second analysis windows.

The following parameters were used:

Sampling frequency: 250 Hz

Window duration: 5 seconds

Segment length: 1250 samples

Window overlap: 1125 samples

Step size: 0.5 seconds

2.5 Alpha-Band Power Extraction

Alpha-band power was calculated within the frequency range of 8–12 Hz.

The filtered EEG signals were divided into non-overlapping five-second windows. Welch's method was applied to each window, and the power within the alpha frequency range was integrated to obtain alpha-band power. Frequency-domain analysis is widely used to examine oscillatory activity in neural signals and to quantify the power associated with specific frequency bands (Cohen, 2014).

2.6 Statistical and Descriptive Analysis

For each EEG channel, the following descriptive measurements were calculated:

- Mean alpha power
- Standard deviation of alpha power
- Number of analysed windows

The results from the eyes-open and eyes-closed conditions were compared across all eight channels.

2.7 Data Visualisation and Result Storage

Graphs were generated using Matplotlib to visualise the EEG signals, power spectral density, and alpha-power comparisons.

The calculated results were saved as CSV files for future reference and inclusion in the final project report.

3 | RESULTS

3.1 Alpha Power Comparison

After preprocessing the EEG signals and calculating alpha-band power, the results from the eyes-open and eyes-closed conditions were compared across all eight EEG channels. Alpha-frequency oscillations are associated with multiple aspects of brain function, including the coordination and regulation of neural activity, indicating that changes in alpha power may reflect differences in functional brain states (Palva & Palva, 2007).

The mean alpha power was calculated for each channel under both recording conditions.

CHANNEL	Mean Alpha Power — Eyes Open (μV²)	Mean Alpha Power — Eyes Closed (μV²)
EXG 1	7.4275	88.2761
EXG 2	7.5614	62.5520
EXG 3	7.4387	76.5330
EXG 4	4.1368	40.7754
EXG 5	6.9118	51.6177
EXG 6	4.5859	34.2347
EXG 7	5.9465	43.5447
EXG 8	2.797	42.383

Table 3. Mean Alpha Power Comparison

3.2 Statistical Summary

The mean and standard deviation of alpha power were calculated using the analysed five-second windows.

There were 60 windows in the eyes-open condition and 61 windows in the eyes-closed condition for each channel.

The standard deviation was higher during eyes-closed conditions across all eight channels, indicating greater variation in alpha power across the analysed windows.

Channel	SD — Eyes Open (μV²)	SD — Eyes Closed (μV²)
EXG 1	2.9898	35.7978
EXG 2	3.1692	23.5732
EXG 3	2.8006	31.2785
EXG 4	1.5938	17.2140
EXG 5	2.8274	22.4915
EXG 6	2.2414	15.4559
EXG 7	3.3905	17.9520
EXG 8	0.9938	13.3099

Table 4.Standard Deviation of Alpha Power

3.3 Alpha Power Difference

The difference in mean alpha power between the two recording conditions was calculated by subtracting the eyes-open mean alpha power from the eyes-closed mean alpha power for each EEG channel.

The results showed a positive difference across all eight EEG channels, indicating that mean alpha power was higher during the eyes-closed condition.

The largest difference was observed in EXG 1, while the smallest difference was observed in EXG 6.

EEG Channel	Eyes Closed — Eyes Open (μV²)
EXG 1	80.8487
EXG 2	54.9906
EXG 3	69.0943
EXG 4	36.6386
EXG 5	44.7058
EXG 6	29.6487
EXG 7	37.5983
EXG 8	39.5860

Table 5.Difference in Mean Alpha Power

The results show that mean alpha power was higher during the eyes-closed condition than during the eyes-open condition across all eight EEG channels.

EXG 1 showed the largest difference in mean alpha power, while EXG 6 showed the smallest difference.

This observation is consistent with the commonly reported increase in alpha activity during eyes-closed resting conditions.

The results represent a descriptive comparison of the two recordings and do not establish statistical significance across a wider population.

3.4 EEG Signal Visualization

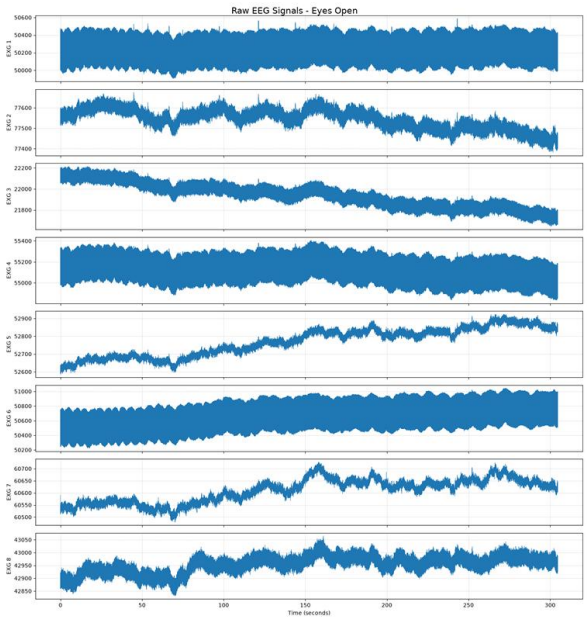


Figure 1. Raw EEG signals during the eyes-open condition across eight EEG channels (EXG 1–EXG 8).

Figure 1 shows the raw EEG signals recorded during the eyes-open condition. The signals were visualized before preprocessing to examine the original EEG activity across all eight channels.

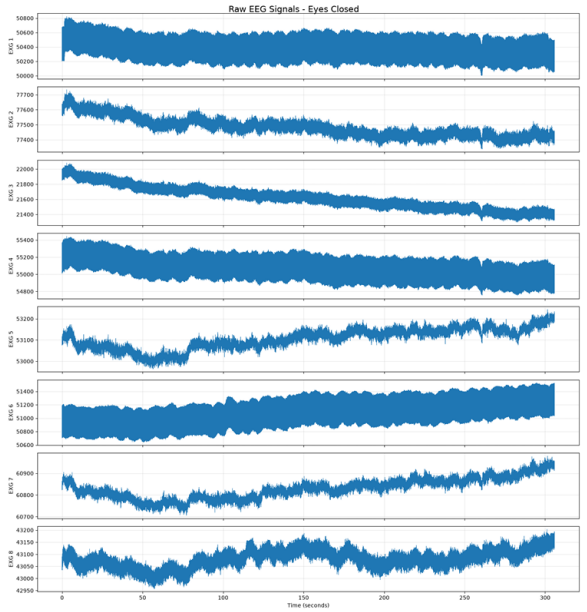


Figure 2. Raw EEG signals recorded during the eyes-closed condition across eight EEG channels (EXG 1–EXG 8).

Figure 2 shows the raw EEG signals recorded during the eyes-closed condition before preprocessing. The graph presents the EEG activity across all eight channels and allows the original signals to be visually examined before further analysis.

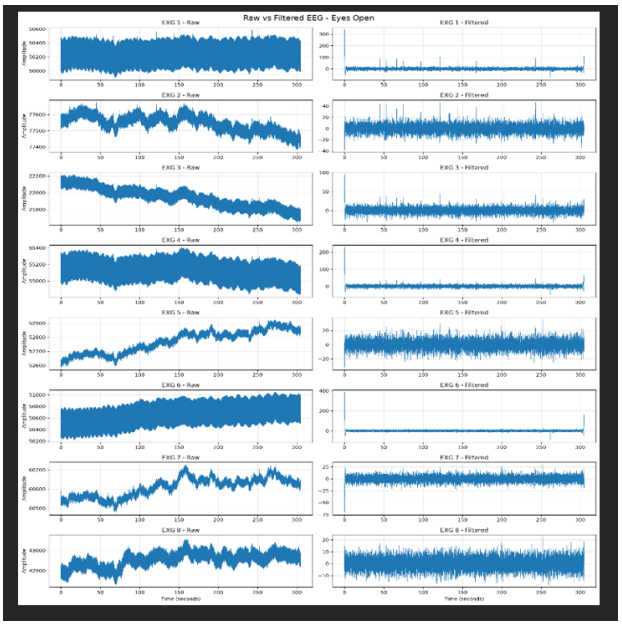


Figure 3. Raw vs Filtered EEG – Eyes Open

Figure 3 Raw versus filtered EEG signals during the eyes-open condition across eight EEG channels (EXG 1–EXG 8). The left column shows the raw EEG signals, while the right column shows the signals after band-pass (0.5–40 Hz) and notch (50 Hz) filtering.

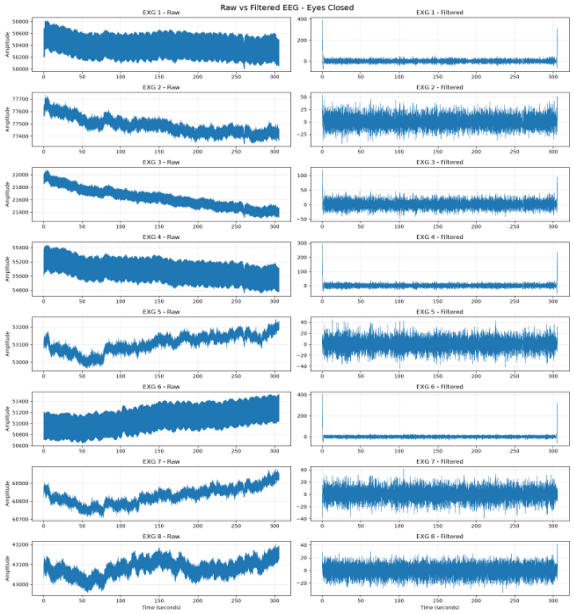


Figure 4. Raw vs Filtered EEG – Eyes Closed

Figure 4 Raw versus filtered EEG signals during the eyes-closed condition across eight EEG channels (EXG 1–EXG 8). The left column shows the raw EEG signals, while the right column shows the signals after band-pass (0.5–40 Hz) and notch (50 Hz) filtering.

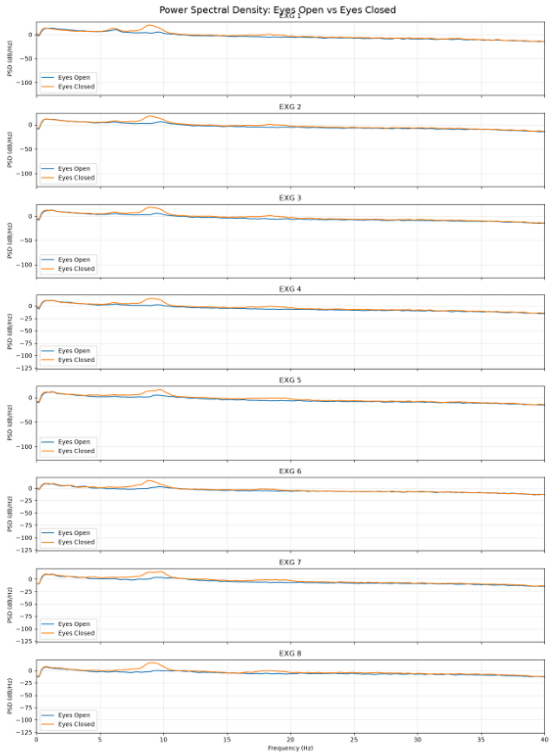


Figure 5.Power Spectral Density (PSD) Comparison—EXG 1

Figure 5 Comparison of power spectral density (PSD) for EEG channel EXG 1 during eyes-open and eyes-closed conditions. The graph illustrates differences in spectral power across frequencies, including the alpha-frequency band (8–12 Hz).

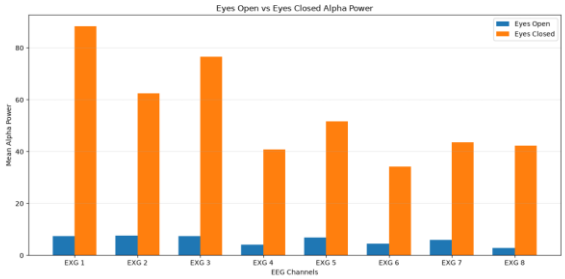


Figure 6. Alpha Power Comparison — Eyes Open vs Eyes Closed

Figure 6 Comparison of mean alpha-band power (8–12 Hz) between eyes-open and eyes-closed conditions across eight EEG channels (EXG 1–EXG 8). Alpha power was higher during the eyes-closed condition across all eight channels.

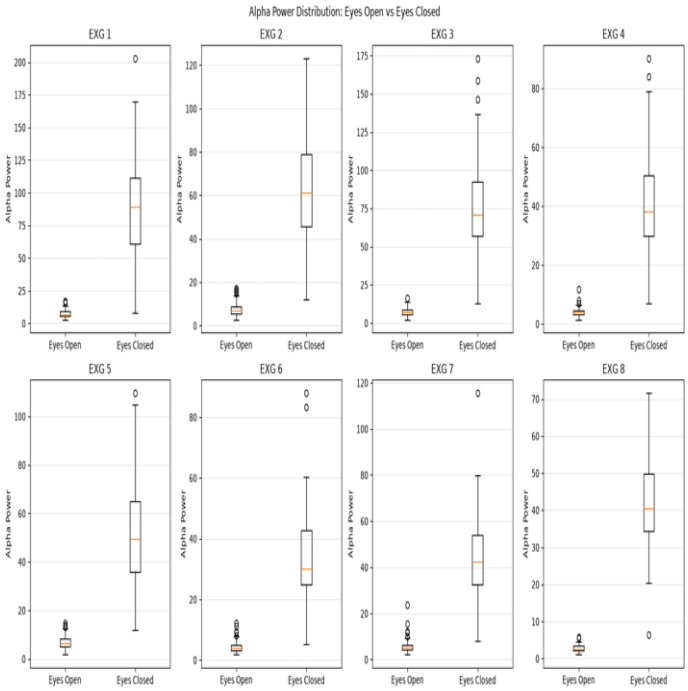


Figure 7. Alpha Power Distribution — Eyes Open vs Eyes Closed

Figure 7 Distribution of alpha-band power (8–12 Hz) across the analysed five-second EEG windows during eyes-open and eyes-closed conditions. The distribution illustrates differences in alpha-power levels and variability between the two recording conditions.

4 | DISCUSSION

The present study examined differences in alpha-band (8–12 Hz) EEG power between eyes-open and eyes-closed resting conditions using eight EEG channels recorded with the Unicorn BCI Core-8 system. The results showed consistently higher mean alpha power during the eyes-closed condition across all eight channels.

The largest mean alpha power during eyes closure was observed in EXG 1 (88.2761 μV^2), followed by EXG 3 (76.5330 μV^2) and EXG 2 (62.5520 μV^2). In contrast, mean alpha power during the eyes-open condition ranged from 2.7970 μV^2 to 7.5614 μV^2. These findings demonstrate a clear descriptive difference in alpha-band activity between the two recording conditions.

The increase in alpha power during eye closure is consistent with the established association between alpha oscillations and reduced visual sensory processing during relaxed wakefulness. Opening the eyes introduces visual input and is commonly associated with a reduction in alpha activity, often described as alpha suppression or desynchronization (Foxe & Snyder, 2011). Alpha-band oscillations have also been associated with the regulation of neural information processing, particularly through mechanisms of functional inhibition (Jensen & Mazaheri, 2010). Previous research has also linked alpha-band activity over occipital regions with visual attention and visual target detection, highlighting the functional relevance of alpha oscillations in visual processing (Thut et al., 2006).

Research has also demonstrated that visual conditions can modulate resting-state alpha oscillations, providing further context for the differences in alpha power observed between eyes-open and eyes-closed recordings (Webster & Ro, 2020).

The analysis also revealed greater standard deviations of alpha power during the eyes-closed condition across all eight channels. This suggests greater variability in alpha power across the analysed five-second windows. Such variability may reflect fluctuations in resting-state brain activity over the recording period.

The power spectral density comparison and alpha-power visualizations further supported the descriptive findings. Together, these results demonstrate that the EEG recordings and preprocessing pipeline could capture condition-related differences in alpha-band activity. Reductions in rhythmic EEG power can be described in terms of desynchronization, a concept used to interpret changes in the synchronization of neural oscillatory activity (Pfurtscheller & Lopes da Silva, 1999).

However, the findings should be interpreted as descriptive rather than statistically significant, because no inferential statistical test was performed. The results therefore support the study's objective of comparing resting-state alpha activity between eyes-open and eyes-closed conditions without establishing the statistical significance of the observed differences.

5 | LIMITATIONS OF THE STUDY

The present study has several limitations that should be considered when interpreting the findings.

First, the analysis was based on EEG recordings from the available dataset. Consequently, the results may not be generalizable to a wider population or different recording environments.

Second, the analysis focused specifically on alpha-band activity (8–12 Hz). Other EEG frequency bands, including delta, theta, beta, and gamma, were not examined in detail.

Third, although band-pass and notch filtering were applied to reduce unwanted signal components, residual artifacts associated with eye movements, muscle activity, or other sources of interference may have influenced the EEG measurements. EEG recordings may contain artifacts that affect signal quality, and automated artifact-rejection approaches have been developed to identify and reduce their influence during EEG analysis (Nolan et al., 2010).

Fourth, the statistical analysis was limited to descriptive measures, including mean alpha power and standard deviation. No inferential statistical tests were conducted to determine whether the differences between eyes-open and eyes-closed conditions were statistically significant.

Finally, the analysis used non-overlapping five-second windows to estimate alpha-power variability. The findings may differ if alternative window lengths or spectral estimation parameters are used.

Despite these limitations, the study provides a useful descriptive comparison of alpha-band EEG activity between eyes-open and eyes-closed resting conditions.

6 | CONCLUSION

The present study investigated differences in alpha-band (8–12 Hz) EEG power between eyes-open and eyes-closed resting conditions using recordings obtained from the Unicorn BCI Core-8 system. EEG signals from eight channels were processed using a fourth-order Butterworth band-pass filter (0.5–40 Hz) and a 50 Hz notch filter to reduce unwanted signal components. Power spectral density was estimated using Welch's method, and alpha-band power was calculated and summarized using descriptive statistics.

The results demonstrated consistently higher mean alpha power during the eyes-closed condition across all eight EEG channels. The highest mean alpha power during eyes closure was observed in EXG 1 (88.2761 μV^2), while the eyes-open condition showed comparatively lower alpha-power values. Greater standard deviations were also observed during the eyes-closed condition, indicating increased variability across the analysed five-second windows.

These findings are consistent with the expected increase in resting-state alpha activity during eye closure and demonstrate the usefulness of EEG signal processing and spectral analysis for examining differences in brain activity under different visual conditions.

Although the analysis was descriptive and did not include inferential statistical testing, it successfully demonstrated condition-related differences in alpha-band EEG power. Future research could extend this approach by examining additional EEG frequency bands, incorporating more recordings or participants, and applying appropriate statistical tests to evaluate the reliability of the observed differences.

Overall, the study illustrates how EEG preprocessing, power spectral density estimation, and alpha-band analysis can be combined to investigate resting-state brain activity.

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